PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Angle Preparation A - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt both problems. Related angle/dielectric preparation; no claim that these are the upcoming bonus.

Write all rough calculations and final answers on separate paper. Show your working.

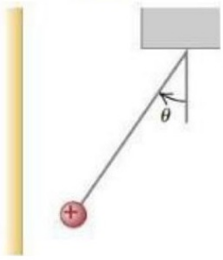
Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Homework 4 - Question 22.33

22.33 •• CP A small sphere with mass 4.00×10^{-6} kg and charge 5.00×10^{-8} C hangs from a thread near a very large, charged insulating sheet (**Fig. P22.33**). The charge density on the surface of the sheet is uniform and equal to -2.50×10^{-9} C/m^2. Find the angle of the thread.

Figure **P22.33**



2. [____] Homework 6 - Question 23.59

23.59 •• CP A small sphere with mass 1.50 g hangs by a thread between two very large parallel vertical plates 5.00 cm apart (**Fig. P23.59**). The plates are insulating and have uniform surface charge densities $+\sigma$ and $-\sigma$. The charge on the sphere is $q = 8.90 \times 10^{-6}$ C. What potential difference between the plates will cause the thread to assume an angle of 30.0° with the vertical?

Figure **P23.59**

